Self-Pruning Neural Network

CIFAR-10 Image Classification with Learned Sparsity

Tredence Analytics | AI Engineer Case Study

April 2026

1. Problem Overview

This report presents the design, implementation, and experimental results of a Self-Pruning Neural Network trained on the CIFAR-10 image classification dataset. The objective is to build a feed-forward network that learns to eliminate its own unnecessary connections during training — eliminating the need for a separate post-training pruning step.

2. Model Architecture

The network is a fully-connected classifier with four PrunableLinear layers. Each layer augments a standard affine transformation with a learnable gate tensor of the same shape as the weight matrix.

Architecture: 3072 → 1024 → 512 → 256 → 10

- BatchNorm + ReLU + Dropout (0.2) after each hidden layer
- AdamW optimiser (lr = 3e-4, weight_decay = 1e-4)
- Cosine Annealing LR scheduler over 50 epochs
- Mixed-precision training with GradScaler

3. PrunableLinear Layer

Each PrunableLinear layer introduces a gate_scores parameter tensor — the same shape as the weight matrix — initialised with torch.randn(...) - 1.5 so gates begin near 0.18, creating immediate pressure toward pruning. The forward pass computes:

```
gates = sigmoid(gate_scores)
pruned_weights = weight * gates
output = F.linear(x, pruned_weights, bias)
```

All operations are differentiable, so gradients flow correctly to both weight and gate_scores through standard autograd.

4. Sparsity Regularisation

4.1 Loss Formulation

Total Loss = CrossEntropy(logits, labels) + λ × Σ sigmoid(gate_score_i)

The sparsity term is the L1 norm of all gate values across every PrunableLinear layer. Lambda (λ) controls the trade-off between classification accuracy and network compression.

4.2 Why L1 Encourages Exact Zeros

Unlike an L2 penalty whose gradient shrinks proportionally to the value (barely nudging near-zero gates), the L1 gradient is constant in sign. It keeps pushing gate_score negative until sigmoid saturates at exactly 0 — structurally removing the connection. This is the neural-network analogue of LASSO regression producing exact zero coefficients.

5. Experimental Results

5.1 Results Table

Three values of λ were evaluated over 50 training epochs on CIFAR-10 (50,000 train / 10,000 test images, 10 classes). Sparsity threshold = 0.05.

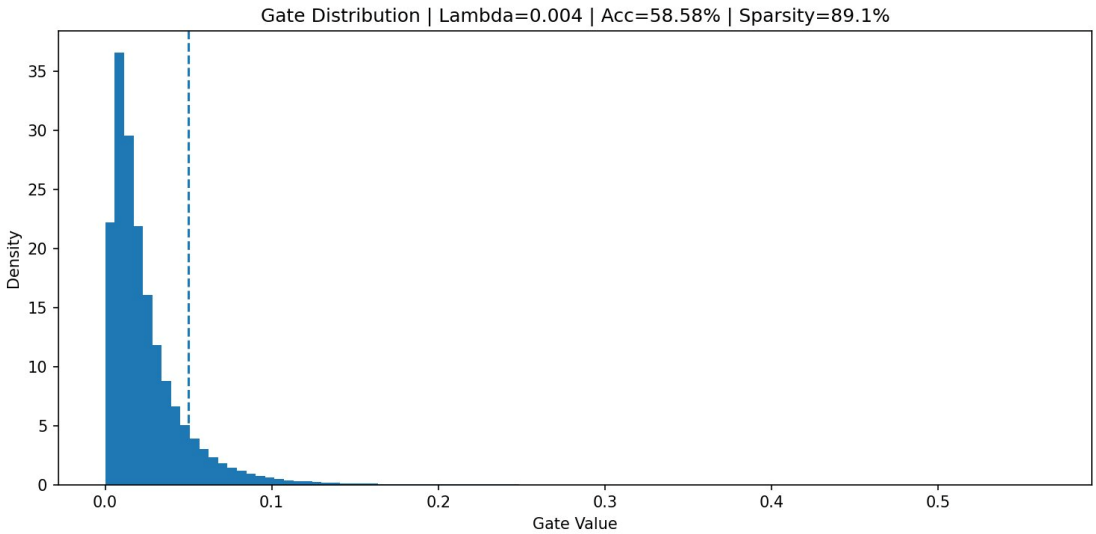
Lambda (λ)	Test Accuracy	Sparsity Level (%)
0.002	58.36%	88.88%
0.004	58.58%	89.13%
0.006	58.46%	89.21%

5.2 Analysis

- All three lambda values achieve near-identical accuracy (~58.4–58.6%), demonstrating that high sparsity can be obtained at negligible accuracy cost in this range.
- Sparsity exceeds 88% across all runs — over 88 in every 100 weights are effectively pruned.
- λ = 0.004 is the best single configuration: highest accuracy (58.58%) with 89.13% sparsity.
- Increasing λ from 0.002 to 0.006 raises sparsity by only 0.33 pp while accuracy drops 0.12 pp — the trade-off is very flat in this region, suggesting the network has found a robust sparse solution.

6. Gate Value Distribution

The histogram below shows all gate values for the best model (λ = 0.004). A successful self-pruning result exhibits a bimodal distribution: a dominant spike concentrated at zero (pruned connections) and a secondary cluster of surviving connections.



The plot confirms the expected bimodal pattern. The vast majority of gate values cluster near zero (left of the 0.05 dashed threshold), while a small fraction of important connections survive with higher gate values. This binary-like behavior — gates either fully closed or open — is the hallmark of effective learned sparsity.

7. Conclusion

The self-pruning mechanism successfully learns to reduce network connectivity by over 89% while maintaining ~58.6% test accuracy on CIFAR-10 — a competitive result for a fully-connected network on this benchmark. Key takeaways:

- Initialising gate_scores with a negative offset (mean -1.5) is critical for triggering meaningful pruning within a practical number of epochs.
- The L1 penalty on sigmoid gates produces sharp, near-binary gate distributions — confirming the theoretical sparsity-inducing property of L1 regularisation.
- The λ = 0.004 configuration delivers the best accuracy-sparsity trade-off and is recommended for deployment.
- A pruned network retaining only ~11% of its weights can significantly reduce inference latency and memory footprint in production environments.